

Rahul Sharma

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Professional Summary

EXPERIENCE

Bilbo.ai, Baltimore, US: AI Engineer

Sept 2025 – May 2026

- Delivered a production healthcare RAG platform for protocol queries, combining hybrid retrieval, reranking, and citation grounding to generate traceable responses with sub-5-second latency.
- Developed eight stage document processing pipeline covering parsing, hierarchy extraction, paragraph-aware chunking, FAISS vector indexing, and quality validation etc, improving citation correctness
- Integrated FAISS and BM25 retrieval with Reciprocal Rank Fusion and cross-encoder reranking, raising top-five relevance by 28% while preserving paragraph-level source attribution.
- Deployed a 4-bit quantized Mistral-7B model through Ollama, implementing prompt engineering, LangGraph stateful orchestration structured output validation, and LLM guardrails
- Designed agentic AI workflows with LangGraph, implementing stateful orchestration, conditional routing, memory, and tool/function calling to automate multi-step healthcare protocol reasoning.

Jet2 and Jet2 Holidays, Leeds, UK: Data Scientist

June 2022 – Aug 2024

- Engineered multi-label intent classification across 1M+ call transcripts by fine-tuning LLMs and distilling a 70B teacher into a 7B model, achieving 94% recall on validated intent labels.
- Developed an anomaly-detection framework for LivePerson chats using topic modelling, moving averages, z-scores, and Isolation Forest, achieving 96% specificity while reducing false-positive operational alerts.
- Benchmarked XGBoost, LSTM, and GRU forecasting models using Optuna hyperparameter tuning and walk-forward validation, reducing MAPE from 18% to 7% against the existing forecasting baseline.
- Deployed forecasting workflows in Dataiku with scheduled retraining, Pydantic-based data validation, and Tableau dashboards, maintaining forecast-error variation below 2% across 18+ months of planning cycles.
- Built transcript summarization pipelines using Snowflake Cortex and few-shot prompting, standardizing customer-conversation summaries across contact-center teams

Fibe (formerly EarlySalary): Data Scientist

Sept 2021 – June 2022

- Architected credit-risk scorecard by integrating CIBIL, Experian, transaction, and behavioral data and screening 1,000+ variables using WOE/IV and monotonic binning, achieving 0.94 ROC-AUC.
- Validated scorecard discrimination and stability using ROC-AUC, KS, Gini, out-of-time testing, and risk-segment analysis, strengthening model robustness across changing borrower populations and lending cohorts.
- Established MLOps workflows using MLflow for experiment tracking, model versioning, metric comparison, and reproducible validation across forecasting and NLP model development cycles.
- Automated feature engineering, model scoring, validation gates, and scheduled production workflows, reducing credit-risk model turnaround 15x from three days to under five hours.

Advanced Technology Consulting Service: Associate Data Scientist

Nov 2020 – Sept 2021

- Engineered distributed PySpark ETL pipelines processing 10M+ multi-source records, reducing production batch runtime from 4 hours to 18 minutes through scalable data transformation and processing.
- Built a configuration driven data quality framework incorporating data validation, profiling and referential integrity, strengthening production data reliability across 4+ enterprise engagements.
- Deployed a fine-tuned Mask R-CNN document-detection model with ResNet-101-FPN on Azure, achieving 92% mAP and reducing manual document-region review by 63%.
- Containerized Python and ML workloads using Docker and automated testing and deployment through CI/CD pipelines, standardizing model releases across development and production environments.

Amazon Pay India: Research Analyst

May 2020 – Nov 2020

- Developed a fraud-risk classification model using behavioural features, customer segmentation, and hypothesis testing, achieving 95% F1-score and strengthening identification of high-risk customer segments.
- Transformed raw KYC JSON data into standardized address records using regex, rule-based entity extraction, and data normalization, increasing usable customer-location coverage by 37% for risk modelling.
- Applied chi-squared tests, mutual information, and LASSO feature selection across transactional variables, reducing model input dimensionality by 42%
- Built Power BI dashboards tracking operating cash flow, credit KPIs, and portfolio risk indicators, reducing risk-reporting turnaround by 30% and supporting executive credit decisions.

EDUCATION

University of Maryland, College Park, US

Aug 2024 - May 2026

Master of Science in Data Science,

Indian Institute of Technology (BHU), Varanasi, India

July 2016 - June 2020

Bachelor of Technology in Engineering,

PROJECTS

Agentic Risk Intelligence Platform (AI Cargo)

Mar 2026 - Apr 2026

- Engineered a six-layer cold-chain risk platform ingesting live shipment telemetry and scoring pharmaceutical spoilage risk in real time using compliance rules and machine-learning signals.
- Combined 14 engineered features and eight deterministic compliance rules with Optuna-tuned XGBoost and SHAP explanations to generate interpretable, auditable risk scores for high-risk shipments.
- Deployed a FastAPI backend with 25 endpoints, WebSockets, React, Supabase, and Vercel, integrating four LLM providers with deterministic fallback for resilient inference routing.

Skills

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|------------------------|--------------------|-----------------|
| • Agentic AI | • Python | • Model Serving |
| • Multi-Agent Systems | • C/C++ | • ETL Pipelines |
| • LangGraph | • Bash | • AWS |
| • LangChain | • Transformers | • Azure |
| • LLM Evaluation | • Vector Databases | • GCP |
| • LLMops | • Semantic Search | • Snowflake |
| • LangSmith | • MLflow | • Databricks |
| • vLLM | • FastAPI | • Spark |
| • Speculative Decoding | • Docker | • SQL |
| • Fine-Tuning | • Kubernetes | |
| • PEFT | • CI/CD | |

ACHIEVEMENTS

- Co-author, IEEE ICSCDS 2025, "Deep Learning for Reliable Routing in WANETs" (peer-reviewed) — Publication. Paper
- Won first prize and \$4,000 in the Smith Agentic AI Challenge by building a pharmaceutical cold-chain platform combining FDA/GDP rules with explainable ML.